

Project summary: best outcomes by mini-project

Higher is better for NLU metrics; lower is better for LM perplexity.

LM/partA

ablation_d_model_192

dev PPL 62.62 / test 53.65

Scratch GPT2 target PPL < 250 met.

LM/partB

rank8_alpha16_qkv

dev PPL 30.11 / test 27.06

Manual LoRA beats scratch with <0.4% trainable params.

NLU/partA

ablation_n_heads_8

dev intent 0.968, slot F1 0.942

Best scratch GPT2 by combined dev score.

NLU/partB

bert_ontology_report

test intent 0.978, slot F1 0.956

Best pretrained NLU model.